Counter AI: What Is It and What Can You Do About It?

Carnegie Mellon University
Software Engineering Institute

Authors

Nathan VanHoudnos

Carol Smith

Matthew Churilla

Shing-Hon Lau

Lauren McIlvenny

Greg Touhill

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AS THE STRATEGIC IMPORTANCE OF AI INCREASES, SO TOO DOES THE IMPORTANCE OF DEFENDING THOSE AI SYSTEMS [NSCAI 2021]. To understand AI defense, it is necessary to understand AI offense—that is, counter AI.

This paper describes counter AI. First, we describe the technologies that compose AI systems (the AI Stack) and how those systems are built in a machine learning operations (MLOps) lifecycle. Second, we describe three kinds of counter-AI attacks across the AI Stack and five threat models detailing when those attacks occur within the MLOps lifecycle.

Finally, based on Carnegie Mellon University (CMU) Software Engineering Institute (SEI) research and practice in counter AI, we give two recommendations:

- In the long-term, the field should invest in AI engineering research that fosters processes, procedures, and mechanisms that prevent vulnerabilities being introduced into AI systems.
- In the near-term, the field should develop the processes necessary to efficiently respond to and mitigate counter-AI attacks, such as building an AI Security Incident Response Team (AISIRT) and extending existing cybersecurity processes like the *Computer Security Incident Response Team (CSIRT) Services Framework* [FIRST 2019].

The AI Stack and MLOps: AI Technologies and AI Processes

The AI Stack is a pictorial representation of the various technologies and areas of research that are necessary to build out a breadth of AI capabilities [Moore 2018]. It is useful for understanding what AI systems are, but it does not depict how AI systems are built.

In this section, we summarize the AI Stack and develop a companion MLOps lifecycle to describe the process by which AI systems are built.

The left side of Figure 1 shows the AI Stack [Moore 2018]. The gray vertical bar, Ethics, spans the horizontal layers of the stack to emphasize the importance of building and using AI systems that align with the core democratic values of the United States. Ethics encompasses efforts known as AI assurance or responsible AI and includes the tools and techniques necessary to build an AI system that is aligned with user needs and ethical principles, meets technical specifications throughout the various stages of the development and acquisition processes, and is continuously monitored during operational or mission use.

The bottom two horizontal layers, Computing and Devices, form the basis of the hardware, software, and perception systems on which AI systems run. Computing in this formulation is broad, encompassing all the systems, networks, programming languages, operating systems, and hardware that enable computation. The Device layer is similarly broad, referencing the sensors and components needed for machines to perceive the world around them, including everything from cameras to light detection and ranging (LIDAR), and from synthetic aperture radar (SAR) to cyber sensors, such as net flow monitors.

The Massive Data Management (red) and Machine Learning (green) layers form the core of the AI system. The Massive Data Management layer includes the selection and analysis of data, preparation of data, and overall data management. Data are ingested and then used to identify statistical patterns in the Machine Learning layer. The Machine Learning layer represents the whole academic field of statistical machine learning, including supervised, unsupervised, and reinforcement learning approaches.

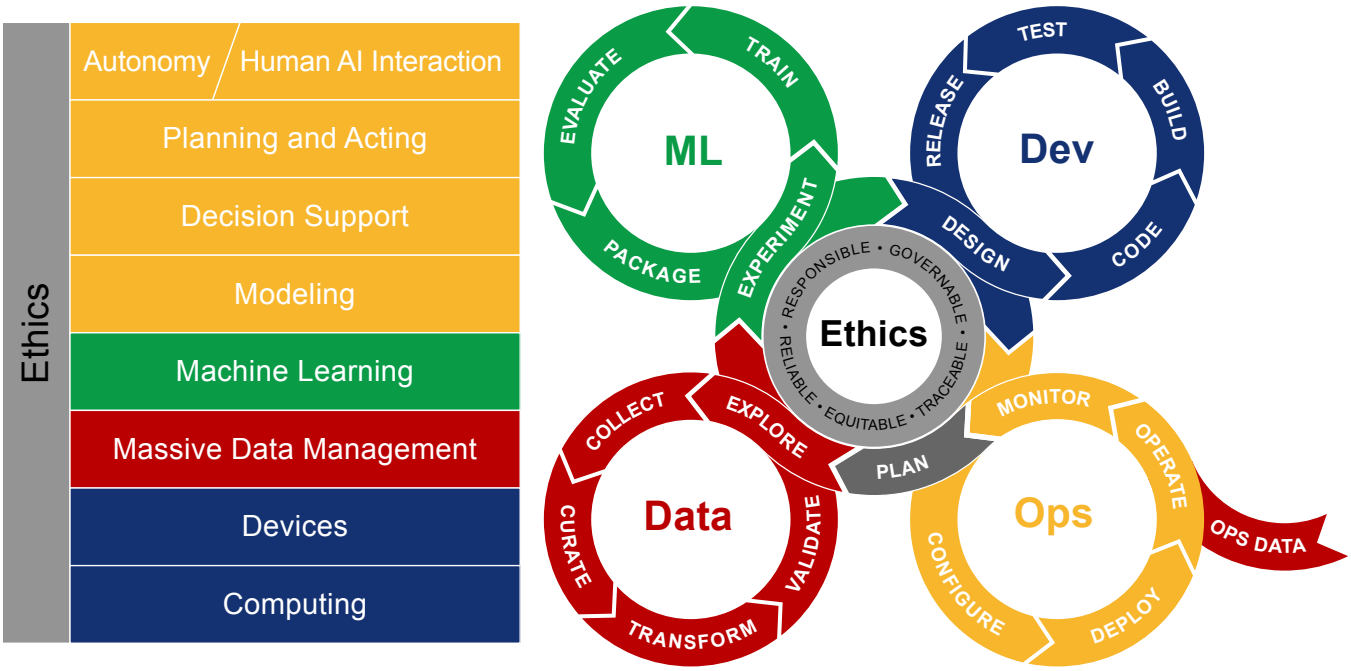

Figure 1: The AI Stack and a Companion MLOps Lifecycle

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The top set of layers in the AI Stack represent the broad capabilities that are often ascribed to AI systems.

- The Modeling layer encompasses structuring of knowledge about the world to synthesize data into higher order concepts.
- The Decision Support layer encompasses the various ways that models synthesized from data can be used to help either humans or algorithms make decisions.
- The Planning and Acting layer represents AI systems that team with humans to make and carry out plans.
- The Autonomy and Human AI Interaction layer represents the spectrum of engagement and actions delegated by a human to an AI system.

In some cases, the AI system displays information in an interface for a human to then act; in others, the AI system may be designed for human-machine teaming scenarios; or the AI system may be designed to act without further guidance, in full autonomy.

The right side of Figure 1 is a companion MLOps lifecycle adapted from an earlier MLOps lifecycle [ML4Devs 2022]. The MLOps lifecycle is centered on the five DoD ethical principles for AI [DoD 2020]. The lifecycle is composed of five connected loops: Ethics, Data, ML (machine learning), Dev (development), and Ops (operations). Ethics (grey) takes a prominent place here (as it did in the AI Stack) because, again, considerations of ethics should permeate the whole process. The MLOps lifecycle starts from Ethics, at Plan (grey arrow), where the CONOPS (concept of operations, including end user needs and context) and measures of performance are defined by the team and stakeholders for the eventual system.

The next loop is Data (red), which is entered when the system is first developed and when new training data are required. Its five components are (1) Explore what data already exist, (2) Collect any new data required, (3) Curate and, if necessary, label the data for a given use case, (4) Transform the data for use in machine learning, and (5) Validate that the data meet the requirements of the system and its users. The next loop is ML (green), which is entered when a new ML model is required, or the model is changed. Its four components are (1) Experiment with various approaches, (2) Train the necessary models, (3) Evaluate the quality attributes of the models, and (4) Package the selected models for use within an AI system or submission to a model zoo, that is, a curated collection of models.

When new features are required, the system moves into Dev for development (blue). Its five components are (1) Design the features, (2) Code the features, (3) Build the new system, (4) Test the system for functionality, and (5) Release a new version of the system. The final loop is Ops, for Operations (yellow), with four components (1) Configure the system, (2) Deploy the system, (3) Operate the system, and (4) Monitor the system. The last three components run continuously while the system is in use. Note that Monitor feeds directly into Plan, implying that as information about the system is collected through monitoring, new Data, ML, and Dev loops can be integrated, as needed, to build an improved version of the system. Further note that the Ops loop explicitly includes an Operational Data input because the machine learning component of an AI system requires data.

Counter AI: Three Attacks and Five Threat Models

In this section, we introduce three kinds of counter-AI attacks across the AI Stack and five threat models describing when those attacks occur within the MLOps lifecycle. Briefly, AI systems are fundamentally insecure because there exist vulnerabilities in each layer of the AI Stack and plausible exploits at each step of the MLOps lifecycle.

Across the AI Stack, there are three kinds of counter-AI attacks: cybersecurity attacks, adversarial machine learning attacks, and adversarial AI attacks. Cybersecurity attacks use either physical or cyber methods to target the Computing and Device layers of the AI Stack. Physical cybersecurity attacks can include deny, degrade, and destroy strategies, such as jamming or destroying sensors in autonomous systems in operations. Cyber-based attacks similarly provide a means to deny, degrade, and, in some cases, destroy the software that operates the AI capability.

Adversarial machine learning (AML) attacks use machine learning methods to target the Massive Data Management and Machine Learning layers of the AI Stack. This class of attacks is uniquely effective because Machine Learning is a key component low in the stack that is both exposed during the Ops portion of the MLOps lifecycle and fundamentally vulnerable. In general, an attacker can counter ML by making it learn the wrong thing, do the wrong thing, or reveal the wrong thing about either the MLOps process that created it or the properties of its training data [Beieler 2019].

Generally, countering AI with AML follows a common three-step pattern:

1. *Model the target AI system.* To attack an AI system, the attacker must either have access to the target AI system within its operational context or be able to gather sufficient information about the target to build a proxy (an approximation of the target that is likely to share the same vulnerabilities as the target during its operations).
2. *Train the counter on the model.* In typical ML, the data are held constant as the model learns. In AML this is reversed: the attacker "trains the data" as the model is held constant. Controlling the input in this way allows the attacker to identify how best to poison a model trained on a given set of data (learn attack), drive the target system to a desired state (do attack), or reveal information about the training data or model (reveal attack).
3. *Test the counter.* In typical ML, a trained model is evaluated against several hold-out data sets in several contexts to provide evidence that the model is generalizable. Similarly in AML, a trained counter is evaluated against several hold-out models in several contexts to determine its efficacy as a learn, do, or reveal attack within the operational context of the target AI system.